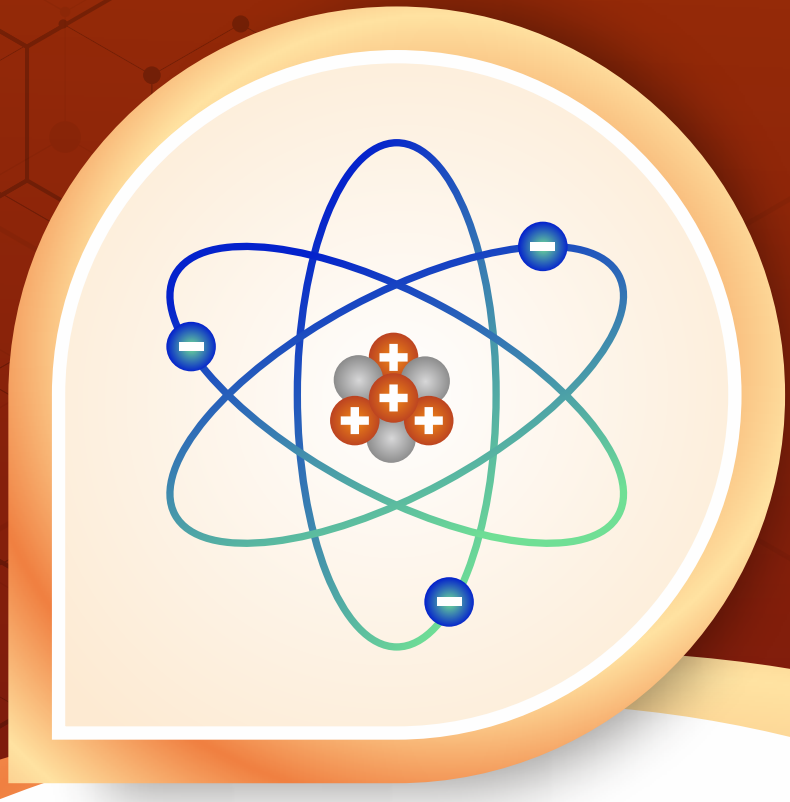
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Atomic Structure

(English Medium)

EXERCISE # S-I

FUNDAMENTAL PARTICLES

1. The ratio of specific charges $\left(\frac{q}{m}\right)$ of an α-particle and a deuteron is x : 1.00. The value of 'x' is.

AS0001

2. Specific charges of two particles A and B are in ratio 2 : 3. If their mass ratio $m_A : m_B$ is 2 : 3, then the ratio of their charges $\left(\frac{q_A}{q_B}\right)$ is 4.00 : x. The value of 'x' is.

AS0002

3. Arrange electron (e), proton (p) , neutron (n) and α-particles (α) , in the increasing order of their e/m ratio.

AS0003

4. In oil drop experiment, the following charges were experimentally determined for five oil droplets 1.20×10^{-18} unit, 3.00×10^{-18} unit, 6.00×10^{-18} unit, 2.40×10^{-18} unit and 3.60×10^{-18} unit (unit is arbitrary). What should be the charge on electron in this arbitrary unit (in the order of 10^{-19}) ?

AS0004

5. The kinetic energy of a charged particle (charge = q) accelerated by a potential difference of 10000 volt is 1.6×10^{-15} Joule. The value of 'q' (in 10^{-19} coulomb) is.

AS0005

RUTHERFORD'S ATOMIC MODEL

6. If the radius of an atom is 2Å and the radius of its nucleus is 4×10^{-15} m. The ratio of volume of nucleus to that of atom is x : 10^{15}. The value of 'x' is.

AS0006

7. Radius of nucleus may be given as $R_N = R_0.A^{1/3}$, where A = mass number and R_0 = constant. Calculate the density (in 10^{17} kg m^{-3}) of nucleus of an atom if $R_0 = 1.2 \times 10^{-15}$.
[$N_A = 6 \times 10^{23}$, $23 \times 8\pi \times (1.2)^3 = 1000$]

AS0007

8. The ratio of distances of closest approach of a proton and an α-particle projected towards the same nucleus with the same initial kinetic energy is x : 1. The value of 'x' is.

AS0008

9. An α-particle of K.E. 5.4 MeV is projected towards Cr-nucleus (Z = 24). What is its distance of closest approach (in 10^{-14} m) ? ($e = 1.6 \times 10^{-19}$C)

AS0009

10. With what speed (in 10^7 ms^{-1}) an α-particle should be projected towards stationary krypton nucleus (Z = 36) such that it approaches upto 2.4×10^{-14}m from the nucleus ?
($N_A = 6 \times 10^{23}$, $e = 1.6 \times 10^{-19}$C).

AS0010

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WAVE CHARACTERISTICS

11. Calculate the wave length (in nm) of an electromagnetic radiation of frequency 2×10^{15} Hz.

AS0011

12. A radiostation emits the radiations of wavelength 2000 m (meter band of station). What is the frequency band of that station, in kHz (Frequency of emitted radio waves) ?

AS0012

13. The wave numbers of two electromagnetic radiations are 4×10^6 m^{-1} and 2×10^5 cm^{-1}. The ratio of their frequencies is x : 1.00. the value of 'x' is.

AS0013

14. An electromagnetic radiation makes 5000 waves in 20 cm. The frequency of radiation (in 10^{12} Hz) is.

AS0014

PLANCK'S QUANTUM THEORY

15. Calculate the energy (in 10^{-19} Joule) of a photon of wave length 4000 Å.

AS0015

16. The eyes of certain member of the reptile family pass a single visual signal to the brain when the visual receptors are struck by photons of wavelength 662 nm . If a total energy of 3.0×10^{-18} J is required to trip the signal, what is the minimum number of photons that must strike the receptor.
($h = 6.62 \times 10^{-34}$ J × s)

AS0016

17. Find the number of photons (in the order of 10^{22}) of radiation of frequency 5×10^{13} s^{-1} that must be absorbed in order to melt one gm ice when the latent heat of fusion of ice is 330 J/g.
($h = 6.6 \times 10^{-34}$ J × sec.)

AS0017

18. A certain dye absorbs 4000 Å and fluoresces at 5000 Å these being wavelengths of maximum absorption that under given conditions 40% of the absorbed energy is emitted. If the ratio of the number of quanta emitted to the number absorbed is x : 1.00, the value of 'x' is.

AS0018

19. The reaction between H_2 and Br_2 to form HBr in presence of light is initiated by the photo decomposition of Br_2 into free Br atoms (free radicals) by absorption of light. The bond dissociation energy of Br_2 is 192 KJ/mole. What is the longest wavelength (in nm) of the photon that would initiate the reaction.
$\left(1\frac{eV}{atom} = 96kJ/mol\ ;\ hc = 1240\ ev \times nm\right)$

AS0019

20. The quantum yield for decomposition of HI is 0.2. In an experiment 0.01 moles of HI are decomposed. Find the number of photons (in the order of 10^{22}) absorbed. ($N_A = 6 \times 10^{23}$)

AS0020

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PHOTOELECTRIC EFFECT

21. The minimum energy required for the emission of a metal electron is 1.324×10^{-18} J. Calculate the threshold wavelength (in nm) required to eject the electron ?

AS0021

22. A light source of wavelength λ illuminates a metal and ejects photo electron with $(KE)_{max}$ = 1 eV. Another light source of wave length $\frac{\lambda}{3}$, ejects photoelectrons from same metal with $(KE)_{max}$=5eV. Find the value of work function (eV) of metal.

AS0022

23. The dissociation energy of H_2 is 482.5 KJ/mol. If H_2 is exposed to radiant energy of wavelength 124 nm, what % of radiant energy will be converted into K.E.

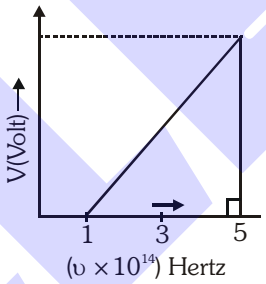
$(1\frac{ev}{atom} = 96.5$ kJ/mol, hc = 1240 eV × nm)

AS0023

24. The K.E. of an electron emitted from tungstan surface is 3.06 eV. What voltage (in volt) would be required to bring the electron to rest.

AS0024

25. Stopping potential [V volts] is plotted against frequency of light used [υ].
Find work function (eV). (h = 6.62×10^{-34} J × s)



AS0025

BOHR'S MODEL

26. If the mass of electron is doubled, then find the new radius (in Å) of 1st orbit of H-atom ?

AS0026

27. If an electron in H-atom jumps from one orbit to other its angular momentum doubles. If the new radius is 'x' times the old radius, then find x ?

AS0027

28. The radius of the an orbit of hydrogen atom is 0.85 nm. Calculate the speed (in 10^5 m/sec) of electron in this orbit.

AS0028

29. The velocity of e^- in a certain Bohr orbit of the hydrogen atom bears the ratio 1:275 to the speed of light.

(i) What is the quantum no. "n" of the orbit.

(ii) Calculate the wave number (in 10^6 m^{-1}) of the radiation emitted for the transition from the quatum state (n+1) to the ground state.

AS0029

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30. If the average life time of an excited state of H atom is of order 10^{-8} sec, estimate how many revolutions (in the order of 10^6) an e^- makes when it is in the state n = 2 and before it suffers a transition to n =1 state.

AS0030

31. Calculate the frequency of revolution of electron (in 10^{14} revolutions per sec) in the first Bohr orbit in a H-atom.

AS0031

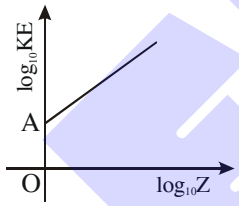
32. Calculate energy (in eV) of electron which is moving in the orbit of a hydrogen atom that has its radius, sixteen times the radius of first Bohr orbit for H–atom.

AS0032

33. The energy of electron in an excited H-atom is –3.4 eV. Calculate angular momentum of electron (in 10^{-34} kg m^2 s^{-1}) in the given orbit.

AS0033

34. In a H-like atom for an electron revolving in n^{th} orbit, the variation of $\mathbf{\log_{10} KE}$ versus $\mathbf{\log_{10} Z}$ is plotted as following -



where Z is atomic number & KE is kinetic energy of electron (in eV). If OA = $\log_{10}$ 3.4 then find the value of principle quantum number of the electron.

AS0034

35. The energy of the second stationary state in Li^{+2} ion is -4.95×10^{-18} J. Find ionization energy (in KJ/mole) for He^+ ions in its ground state ($N_A = 6 \times 10^{23}$).

AS0035

36. The energy of an electron in the first orbit of He^+ is -8.72×10^{-18} J. The energy of the electron in the first orbit of hydrogen (in 10^{-18} J) would be :

AS0036

37. A single electron orbits around a stationary nucleus of charge +Ze where Z is atomic number and 'e' is the magnitude of the electric charge. The hydrogen like species required 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find

(i) the value of Z and give the hydrogen like species formed.

(ii) the kinetic energy (in eV) of the electron in the first Bohr orbit.

AS0037

38. A hydrogen like atom with atomic number 'Z' is in higher excited state of quantum number 'n'.This xcited state atom can make a transition to the first excited state by successively emitting two photons of energies 10.2 eV and 17.0 eV respectivley. Alternatively, the atom from the same excited state can make a transition to the 2nd excited state by emitting two photons of energies 4.25 eV and 5.95 eV respectively. Calculate the value of 'Z'.

AS0038

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39. H-atom is exposed to electromagnetic radiation of 1026 Å and gives out induced radiations (radiations emitted when e^- returns to ground state).Calculate λ (in Å) of induced radiations.

AS0039

40. At what minimum atomic number a transition from n=2 to n=1 energy level would result in the emission of X–rays with $\lambda = 3.0 \times 10^{-8}$ m?

AS0040

41. A doubly ionised lithium atom is hydrogen like with atomic number z = 3. Find the wavelength (in Å) of the radiation required to excite the electron in Li^{2+} from the first to the third Bohr orbit.

$\left(\frac{1}{R} = 912\,Å\right)$

AS0041

42. If shortest wavelength of H-atom in Balmer series is xÅ then,

(i) The shortest wave length in Lyman series is $\frac{x}{y}$ Å. The value of 'y' is.

(ii) The longest wave length in Paschen series is $\frac{x}{z}$ Å. The value of 'z' is.

AS0042

43. Ionization energy of H-atom is 13.6 eV. If H-atoms in the ground state are excited by monochromatic light of energy 12.1 eV, then the maximum number of spectral lines emitted according to Bohr's theory will be -

AS0043

44. Find the number of spectral lines in Paschen series emitted by atomic hydrogen, when electron excited from ground state to 7th energy level, returns back.

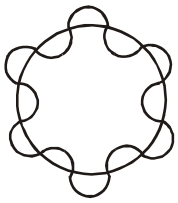
AS0044

45. A stationary He^+ ion emitted a photon corresponding to a first line of the Lyman series. The photon liberated a photoelectron from a stationary hydrogen atom in ground state. What is the speed (in 10^8 cm/sec.) of photoelectron. (Ionisation energy of hydrogen atom = 13.6 eV, $e = 1.6 \times 10^{-19}$ C, Mass of electron = 9.1×10^{-31} kg)

AS0045

DE-BROGLIE AND HEISENBERG EQUATION

46. The figure shows a sample of H-atoms having electron revolving in higher orbit 'n'.



If this electron makes transition from this orbit 'n' to ground state, number of paschen lines emtted are.

AS0046

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47. deBroglie wavelength associated with an electron in 4th orbit of hydrogen atom is $a \times (\pi r_0)$ where r_0 is radius of 1st orbit of hydrogen atom, find value of 'a'.

AS0047

48. The minimum number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number 3 will be -

AS0048

49. Calculate the de-broglie wavelength (in 10^{-66} m) associated with motion of earth (mass 6×10^{24} Kg) orbiting around the sun at a speed of 3×10^6 m/s.

AS0049

50. What is de Broglie wavelength (in pm) associated with an e^- accelerated through potential difference = 15 KV. (Answer to the nearest integer)

AS0050

51. To what effective potential (in volt) a proton beam be subjected to give its protons a wavelength of 1×10^{-10} m. **(Mass of proton = 1.672×10^{-27} kg, e = 1.6×10^{-19} C, h = 6.626×10^{-34} Js)**

AS0051

52. A proton is accelerated to one- tenth of the speed of light. If its velocity can be measured with a precision $\pm$ 1%. What must be its minimum uncertainity in position (in 10^{-12} m).

(Use : $\frac{h}{4\pi}$ = 5.28×10^{-35}) (Mass of proton = 1.6×10^{-24} gm)

AS0052

53. The uncertainty in the position (in cm) of an electron (mass 9.1×10^{-28}gm) moving with a speed of 3×10^4 cm sec^{-1}, uncertainity in speed is 0.011% will be : **(h = 6.6×10^{-34} J × s)**

AS0053

54. If uncertainities in measurement of position and momentum of an electron are equal, then the uncertainity in measurement of its speed is $x \times 10^{12}$. The value of x (the closest nearest integer value) is $m_e = 9.1 \times 10^{-31}$ kg, and h = 6.625×10^{-34} Js.

AS0054

55. With the help of Heisenberg's uncertainty principle, show that electron can never exist in the nucleus.

AS0055

SCHRODINGER EQUATION

56. Calculate the longest distance (in Å) of spherical nodes for '3s' orbital from nucleus in hydrogen atom?

$$R_{3s} = \frac{1}{9\sqrt{3}a_0^{3/2}} (6 - 6\sigma + \sigma^2)\, e^{-\frac{\sigma}{2}} \qquad \text{where } \sigma = \frac{2r}{na_0}$$

AS0056

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57. How many non-spherical subshell are possible that have atleast two maxima if a curve is plotted between radial probability distribution function versus radial distance for which principal quantum number : $n \leq 4$?

AS0057

58. The ratio of angular nodes of 3d orbital and radial nodes of 3p orbital is x : 1 what is the value of x.

AS0058

59. For an orbital in B^{+4} radial function is :

$$R(r) = \frac{1}{9\sqrt{6}}\left(\frac{z}{a_0}\right)^{\frac{3}{2}} (4-\sigma)\,\sigma\, e^{-\sigma/2}$$

where $\sigma = \frac{2Zr}{na_0}$ and $a_0 = 0.529$ Å ; Z = atomic number, r = radial distance from nucleus.

The radial node of orbital is at distance ..... Å from nucleus.

AS0059

60. Find the total number of nodes in 5f-orbital.

AS0060

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EXERCISE # S-II

1. 'α particle' of 3.6 MeV are fired towards nucleus ${}^{A}_{Z}X$. At point of closest separation, distance between 'α particle' and 'X' is 1.6×10^{-14}m. Calculate atomic number of 'X'
[Given : $1/4\pi\varepsilon_0 = 9 \times 10^9$ in S.I. units]

AS0061

2. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{ke^2}{3r^3}$. Use Bohr's theory to obtain energy of such a hypothetical atom.

AS0062

3. In the Bohr's model, for unielectronic species following symbols are used
$r_{n,z} \longrightarrow$ Radius of nth orbit with atomic number "z"
$U_{n,z} \longrightarrow$ Potential energy of electron in n^{th} orbit with atomic number "z"
$K_{n,z} \longrightarrow$ Kinetic energy of electron in n^{th} orbit with atomic number "z"
$v_{n,z} \longrightarrow$ Velocity of electron in n^{th} orbit with atomic number "z"
$T_{n,z} \longrightarrow$ Time period of revolution of electron in n^{th} orbit with atomic number "z"
Calculate z in all in cases.
(i) $U_{1,2} : K_{1,z} = -8 : 1$ (ii) $r_{1,z} : r_{2,1} = 1 : 8$
(iii) $v_{1,z} : v_{3,1} = 9 : 1$ (iv) $T_{1,2} : T_{2,z} = 9 : 32$

AS0063

4. In a hypothetical H-atom, the mass of electron & its charge is double of what we consider, then calculate the total energy (in eV) of electron in the I^{st} orbit of such a hypothetical H-atom ?
[Assuming all others concepts and parameter to be same as we considered in Bohr's model.]

AS0064

5. In H-spectrum, longest wavelength of Lyman is 120 nm and shortest wavelength of Balmer is 360 nm. From this data, find longest wavelength (in nm) of photon, that can ionize this H-atom

AS0065

6. A proton and an electron, both at rest initially, combine to form a H-atom in ground state. A single photon is emitted in this process. Find the wavelength (in nm) of this photon
(use : hc = 1237.6 ev × nm)

AS0066

7. A H-like ion is observed to emit six different wavelengths originating from all possible transitions between a group of levels. These levels have energies between –0.85 eV (Min) and –0.544 eV (Max).
(a) Find atomic number (z) of element
(b) Find quantum numbers of levels between which transitions occur.
(c) Calculate the energy (eV) of the longest wavelength emitted in transitons between the levels.

AS0067

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8. The angular momentum of an electron in a Bohr's orbit of H-atom is 3.1652×10^{-34} kg-m^2/sec. Calculate the wavenumber (in 10^6 m^{-1}) of the spectral line emitted when an electron falls from this level to the ground state. **[h = 6.626 × 10^{-34} Js, R = 1.09 × 10^7 m^{-1}]**

AS0068

9. The ionisation energy of the hydrogen atom is given to be 13.6 eV. A photon falls on a hydrogen atom which is initially in the ground state and excites it to the (n = 4) state. Calculate the wavelength (in Å) of the photon (hc = 12400 Å.eV).

AS0069

10. The ionisation energy of a H-like Bohr atom is 4 Rydbergs.

(i) What is the wavelength (in Å) of radiation emitted when the electron jumps from the first excited state to the ground state.

(ii) What is the radius (in Å) of first Bohr orbit for this atom.
[1 Rydberg = 2.20 × 10^{-18} J, h = 6.60 × 10^{-34} J.s]

AS0070

11. Photon having wavelength 12.4 nm was allowed to strike a metal plate having work function 25 eV. Calculate the

(a) Maximum kinetic energy (in eV) of photoelectrons emitted in eV.

(b) Wavelength (in Å) of electron with maximum kinetic energy in Å.

[Given : hc = 1240 nm.eV]

AS0071

12. The diameter of a dust particle of mass 10^{-3} gm is 2Å. If uncertainity in speed of this particle is $\frac{3.31}{\pi}\times10^{-3}\,m/s$, then find minimum uncertaintity in its position (in 10^{-25} m)

(h = 6.62 × 10^{-34} J.s.)

AS0072

13. The vapours of Hg absorb some electrons accelerated by a potential difference of 4.5 volt as a result of which light is emitted. If the full energy of single incident electron is supposed to be converted into light emitted by single Hg atom, find the wave number (in 10^7 m^{-1}) of the light.

AS0073

14. A photon having λ = 960 Å causes the ionization of a nitrogen atom. Calculate the I.E. per mole of nitrogen atom (in KJ). (Use : hc = 1240 ev × nm and $1\frac{ev}{atom}$ = 96.0 kJ/mol).

AS0074

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15. Mr. Santa has to decode a number "ABCDEF" where each alphabet is represented by a single digit. Suppose an orbital whose radial wave function is represented as

$$\Psi_{(r)} = k_1 \cdot e^{-r/k_2} \quad (r^2 - 5k_3 r + 6k_3^2)$$

From the following information given about each alphabet then write down the answers in the form of "ABCDEF", for above orbital.

Info A = Value of n where "n" is principal quantum number

Info B = No. of angular nodes

Info C = Azimuthal quantum number of subshell to orbital belongs

Info D = No. of subshells having energy between (n + 5)s to (n + 5)p where n is principal quantum number

Info E = Orbital angular momentum of given orbital.

Info F = Radial distance of the spherical node which is farthest from the nucleus

(Assuming $k_3 = 1$)

AS0075

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EXERCISE # O-I

1. Anode rays are made up of :

(A) only protons
(B) only nucleus of atom
(C) Positive residue of atoms
(D) only electrons

AS0076

2. When Lithium $\left({}^{7}_{3}\mathrm{Li}\right)$ vapours were filled in discharge tube for anode ray experiment, the anode rays were found to contain only Li^+ ions. Thus the anode ray particle contains :

(A) 1 proton only
(B) 3 proton and 4 neutron only
(C) 3 proton, 4 neutron and 2 electrons
(D) 3 proton, 3 neutrons and 3 electrons

AS0077

3. The specific charge is maximum for

(A) Na^+ (B) Al^{+3} (C) H^+ (D) Mg^{+2}

AS0078

4. Which of the following particle is not deflected in the magnetic field

(A) Electron (B) proton (C) Neutron (D) Deuteron

AS0079

5. e/m ratio of a particle of charge 2 unit and mass 4 amu is

(A) 4.8×10^7 C/kg (B) 0.5 C/kg (C) 4.8×10^4 C/kg (D) 8×10^{-20} C/kg

AS0080

6. α-particles are projected towards the nucleus of following metals, with the same kinetic energy. The distance of closest approach will be minimum for which metal?

(A) Cu (B) Ag (C) Au (D) Ca

AS0081

7. Which of the following is not a conclusion of Rutherford's atomic model -

(A) Most of the part inside an atom is empty
(B) Almost all mass of an atom is concentrated in the nucleus.
(C) The size of nucleus is very small in compartison to the size of atom
(D) Electron revolves around the nucleus in definite orbits.

AS0082

8. In the different experiements, α-particles, proton, deuteron and neutron are projected towards gold nucleus with the same kinetic energy. The distance of closest approach will be minimum for

(A) α-particle (B) proton (C) dueteron (D) neutron

AS0083

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9. The ratio of the energy of a photon of 2000 Å wavelength radiation to that of 4000 Å radiation is
(A) 1 / 4 (B) 4 (C) 1 / 2 (D) 2

AS0084

10. Small packets of light is called
(A) proton (B) quanta (C) photon (D) spectrum

AS0085

11. Which of the following electromagnetic radiation have greater frequency ?
(A) X - rays (B) Ultraviolet rays (C) Radio waves (D) Visible rays

AS0086

12. Two electromagnetic radiations have wave numbers in the ratio 2 : 3. Their energies per quanta will be in the ratio
(A) 3 : 2 (B) 9 : 4 (C) 4 : 9 (D) 2 : 3

AS0087

13. A radio station is emitting the radiations of frequency 2×10^4 Hz. If its frequency is doubled,
(A) wavelength will be doubled (B) energy per quanta will be doubled
(C) wave number will be halved (D) all of these

AS0088

14. A photon of 400 nm is absorbed by a gas molecule and then the molecule re-emits two photons. One re-emitted photon has wavelength 500 nm. Assuming that there is no change in the energy of molecule, the wavelength of second re-emitted photon is
(A) 100 nm (B) 2000 nm (C) -100 nm (D) 900 nm

AS0089

15. A green bulb and a red bulb are emitting the radiations with equal power. The correct relation between numbers of photons emitted by the bulbs per second is
(A) $n_g = n_r$ (B) $n_g < n_r$ (C) $n_g > n_r$ (D) unpredictable

AS0090

16. The threshold wavelength for ejection of electrons from a metal is 330 nm. The work function for the photoelectric emission from the metal is ($h = 6.6 \times 10^{-34}$ J-s)
(A) 1.2×10^{-18}J (B) 6.0×10^{-19} J (C) 1.2×10^{-20} J (D) 6.0×10^{-12} J

AS0091

17. In the emission of photoelectrons, the number of photoelectrons emitted per unit time depends upon
(A) energy of the incident radiation (B) intensity of the incident radiation
(C) frequency of the incident radiation (D) wavelength of the incident radiation

AS0092

18. Radiations of frequency, ν , are incident on a photosensitive metal. The maximum kinetic energy of photoelectrons is 'E'. When the frequency of the incident radiations is doubled, what is the maximum kinetic energy of the photoelectrons ?
(A) 2E (B) E/2 (C) E + hv (D) E - hv

AS0093

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19. Radiation of λ = 155 nm was irradiated on Li (work function = 5 eV) plate. The stopping potential (in V) is _____.

(A) 3 V (B) 8 V (C) 9 V (D) 5 V

AS0094

20. Electromagnetic radiations having λ = 310 Å are subjected to a metal sheet having work function = 12.8 eV. What will be the velocity of photoelectrons with maximum Kinetic Energy..

(A) 0, no emission will occur (B) 2.18×10^6 m/s

(C) $2.18\sqrt{2} \times 10^6$ m/s (D) 8.72×10^6 m/s

AS0095

21. Bohr's model may be applied to

(A) Na^{10+} ion (B) He atom (C) Be^{2+} ion (D) C^{6+} ion

AS0096

22. If the radius of 3rd Bohr's orbit of H is x, then radius of 4th orbit of Li^{2+} ion would be :-

(A) $\frac{27}{16}$x (B) $\frac{16}{27}$x (C) $\frac{9}{16}$ (D) None of these

AS0097

23. What would be the approximate quantum number, n, for a circular orbit of hydrogen, 1×10^{-5} cm in diameter ?

(A) 31 (B) 43 (C) 40 (D) 39

AS0098

24. If the mass of electron is doubled, the radius of first orbit of H-atom become about

(A) 0.529 Å (B) 0.265 Å (C) 1.058 Å (D) 0.32 Å

AS0099

25. The speed of electron revolving in the 4th orbit of a hydrogen like atom or ion is 1094 km/s. The atom or ion is

(A) H (B) He^+ (C) Li^{2+} (D) Be^{3+}

AS0100

26. How much distance an electron revolving in 3rd orbit of He^+ ion will travel in one second

(A) 1.458×10^6 m (B) 3.28×10^6 m

(C) 4.862×10^5 m (D) 2.917×10^6 m

AS0101

27. The ratio of time taken by electron in revolutions round the H-nucleus in 2nd and 3rd orbits, are

(A) 2 : 3 (B) 4 : 8 (C) 8 : 27 (D) 27 : 8

AS0102

28. Which of the following is not a permissible value of angular momentum of electron in H-atom?

(A) 1.5 $\frac{h}{\pi}$ (B) 0.5 $\frac{h}{\pi}$ (C) 1.25 $\frac{h}{\pi}$ (D) all of these

AS0103

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29. Angular momentum for P–shell electron :–

(A) $\frac{3h}{\pi}$ (B) Zero (C) $\frac{\sqrt{2}h}{2\pi}$ (D) None

AS0104

30. Angular momentum in 2nd Bohr orbit of H-atom is x. Then find out angular momentum in Ist excitetd state of Li^{+2} :–

(A) 3x (B) 9x (C) x/2 (D) x

AS0105

31. The orbit from which when electron will jump in other orbit, energy may be absorbed but not emitted out, will be

(A) 1st orbit (B) 2nd orbit (C) 7th orbit (D) infinite orbit

AS0106

32. The potential energy of electron revolving in the ground state of H-atom is

(A) -13.6 eV (B) -6.8 eV (C) -27.2 eV (D) Zero

AS0107

33. If the potential energy (PE) of hydrogen electron is –3.02eV then in which of the following excited level is electron present :-

(A) 1st (B) 2nd (C) 3rd (D) 4th

AS0108

34. A single electron is revolving in orbits around a stationary nucleus (z = 5). The energy required to excite the electron from third to fourth Bohr orbit will be :-

(A) 4.5 eV (B) 8.53 eV (C) 25 eV (D) 16.53 eV

AS0109

35. A photon of energy 12.75 ev is completely absorbed by a hydrogen atom initially in ground state. The principle quantum number of the excited state is

(A) 1 (B) 3 (C) 4 (D) ∞

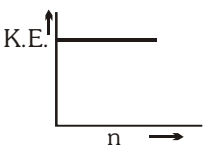
AS0110

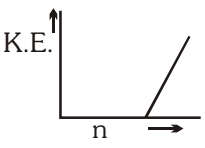
36. An hydrogen atom (ionisation energy 13.6 eV) jumps from third excited state to first excited state. The energy of photon emitted in the process is

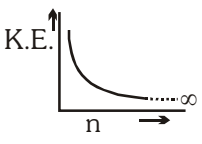
(A) 1.89 eV (B) 2.55 eV (C) 12.09 eV (D) 12.75 eV

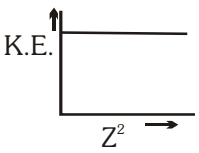
AS0111

37. Which is correct graph :-

(A) 

(B) 

(C) 

(D) 

AS0112

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38. The energy required to remove an electron from the surface of sodium metal is 3.3 eV. What is the longest wavelength of radiation with which it can show photoelectric effect?
Use $h = 6.6 \times 10^{-34}$ J.s
(A) 1.75×10^{-7} m (B) 3.75×10^{-5} m (C) 1.75×10^{-5} m (D) 3.75×10^{-7} m

AS0113

39. Which is correct for any H like species :-
(A) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$ (B) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$
(C) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$ (D) $(E_2 - E_1) = 1/4\ (E_3 - E_2) = 1/9\ (E_4 - E_3)$

AS0114

40. A single electron orbits a stationary nucleus of charge +Ze, where Z is a constant. It requires 47.2 eV to excite electron from second Bohr orbit to third Bohr orbit, find the value of Z :-
(A) 1 (B) 3 (C) 5 (D) 4

AS0115

41. The energy of H–atom in n^{th} orbit is E_n then energy in n^{th} orbit of singly ionised helium atom will be:
(A) $4E_n$ (B) $E_n/4$ (C) $2E_n$ (D) $E_n/2$

AS0116

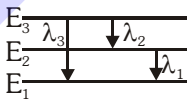
42. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon
(A) 3s (B) 2p (C) 2s (D) 1s

AS0117

43. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen
(A) $5 \rightarrow 3$ (B) $5 \rightarrow 2$ (C) $4 \rightarrow 3$ (D) $4 \rightarrow 2$

AS0118

44. In the following transition which statement is correct



(A) $E_{3-1} = E_{3-2} - E_{2-1}$ (B) $\lambda_3 = \lambda_1 + \lambda_2$
(C) $\nu_3 = \nu_2 + \nu_1$ (D) All of these

AS0119

45. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2$ eV. The same energy change is observed in the second Balmer transition of :-
(A) Li^{2+} (B) Li^{+} (C) He^{+} (D) Be^{3+}

AS0120

46. In a sample of H–atoms, electron transits from 6th orbit to 2nd orbit in multi step. Then total spectral lines (without Balmer series) will be :–
(A) 6 (B) 10 (C) 4 (D) 0

AS0121

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47. What is the shortest wavelength in the Pfund series of He^+ ion :-

(A) $\frac{25}{R}$ (B) $\frac{4}{R}$ (C) $\frac{4R}{25}$ (D) $\frac{25}{4R}$

AS0122

48. The shortest wavelength of He^+ ion in Balmer series is x, then longest wavelength in the Paschene series of Li^{+2} is

(A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$

AS0123

49. The ratio of wave length of photon corresponding to the α-line of Lyman series in H-atom and β-line of Balmer series in He^+ is

(A) 1 : 1 (B) 1 : 2 (C) 1 : 4 (D) 3 : 16

AS0124

50. Three energy levels P, Q, R of a certain atom are such that $E_P < E_Q < E_R$. If λ_1, λ_2 and λ_3 are the wave length of radiation corresponding to transition R → Q ; Q → P and R→ P respectively. The correct relationship between λ_1, λ_2 and λ_3 is

(A) $\lambda_1 + \lambda_2 = \lambda_3$ (B) $\frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ (C) $\lambda_3 = \sqrt{\lambda_1\lambda_2}$ (D) $\frac{2}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

AS0125

51. Number of possible spectral lines which may be emitted in bracket series in H atom, if electrons present in 9th excited level returns to ground level, are

(A) 21 (B) 6 (C) 45 (D) 5

AS0126

52. The value of $(n_2 + n_1)$ and $\left(\mathbf{n_2^2 - n_1^2}\right)$ for He^+ ion in atomic spectrum are 4 and 8 respectively. The wavelength of emitted photon when electron jump from n_2 to n_1 is

(A) $\frac{32}{9} R_H$ (B) $\frac{9}{32} R_H$ (C) $\frac{9}{32 R_H}$ (D) $\frac{32}{9 R_H}$

AS0127

53. An electron, a proton and an alpha particle have kinetic energies of 16E, 4E and E respectively. What is the qualitative order of their de Broglie wavelengths?

(A) $\lambda_e > \lambda_p = \lambda_\alpha$ (B) $\lambda_p = \lambda_\alpha > \lambda_e$ (C) $\lambda_p > \lambda_e > \lambda_\alpha$ (D) $\lambda_\alpha < \lambda_e » \lambda_p$

AS0128

54. The wavelength associated with a golf weighing 200g and moving at a speed of 5m/h is of the order

(A) 10^{-10}m (B) 10^{-20}m (C) 10^{-30}m (D) 10^{-40}m

AS0129

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55. An electron has kinetic energy 2.8×10^{-23} J. de-Broglie wavelength will be nearly :-
($m_e = 9.1 \times 10^{-31}$ kg)
(A) 9.28×10^{-24} m (B) 9.28×10^{-7} m (C) 9.28×10^{-8} m (D) 9.28×10^{-10} m

AS0130

56. What will be de-Broglie wavelength of an electron moving with a velocity of 1.2×10^5 ms^{-1}:-
(A) 6.068×10^{-9} m (B) 3.133×10^{-37} m (C) 6.626×10^{-9} m (D) 6.018×10^{-7} m

AS0131

57. For a valid Bohr orbit, its circumfrence should be:
(A) $= n\lambda$ (B) $= (n-1)\lambda$ (C) $> n\lambda$ (D) $< n\lambda$

AS0132

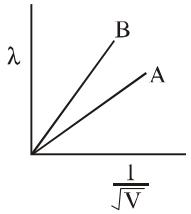
58. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number + 2 :
(A) 3 (B) 4 (C) 2 (D) 1

AS0133

59. How fast is an electron moving if it has a wavelength equal to distance travelled in one second -
(A) $\sqrt{\frac{m}{h}}$ (B) $\sqrt{\frac{h}{m}}$ (C) $\sqrt{\frac{h}{p}}$ (D) $\sqrt{\frac{h}{2\times K.E.}}$

AS0134

60. De-Broglie wavelength of two particles A & B are plotted against $\left(\frac{1}{\sqrt{V}}\right)$; where V is potential on the particles. Which of the following relation is correct about mass of particle (charge is same)
(A) $M_A = M_B$
(B) $M_A > M_B$
(C) $M_A < M_B$
(D) $M_A \leq M_B$



AS0135

61. An α-particle is accelerated through a potential difference of V volts from rest. The de-Broglie's wavelength associated with it is -
(A) $\sqrt{\frac{150}{V}}A^\circ$ (B) $\frac{0.286}{\sqrt{V}}A^\circ$ (C) $\frac{0.101}{\sqrt{V}}A^\circ$ (D) $\frac{0.983}{\sqrt{V}}A^\circ$

AS0136

62. Which quantum number is not related with Schrodinger equation
(A) Principal (B) Azimuthal (C) Magnetic (D) Spin

AS0137

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63. Which is true about ψ :-

(A) ψ represents the probability of finding an electron around the nucleus

(B) ψ represent the amplitude of the electron wave

(C) Both A and B

(D) None of these

AS0138

64. According to Schrodinger model nature of electron in an atom is as :-

(A) Particles only (B) Wave only

(C) Both simultaneously (D) Sometimes waves and sometimes particle

AS0139

65. The orbital angular momentum of an electron in 2s orbital is:

(A) $+\frac{1}{2}.\frac{\hbar}{2\pi}$ (B) Zero (C) $\frac{\hbar}{2\pi}$ (D) $\sqrt{2}.\frac{\hbar}{2\pi}$

AS0140

66. In an excited state, a calcium atom has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 4d^1$. What is the orbital angular momentum for d electron :- $\left(\hbar = \frac{h}{2\pi}\right)$

(A) $\sqrt{4}\hbar$ (B) $\sqrt{16}\hbar$ (C) $\sqrt{6}\hbar$ (D) $\sqrt{10}\hbar$

AS0141

67. Which orbitlal has two angular nodal planes :-

(A) s (B) p (C) d (D) f

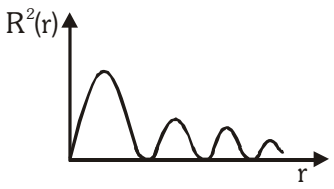
AS0142

68. From the following observations predict the type of orbital :

Observation 1 : x y plane acts as nodal plane

Observation 2 : The angular function of the orbital intersect the three axis at origin only.

Observation 3 : $R^2(r)$ v/s r curve is obtained for the orbital is

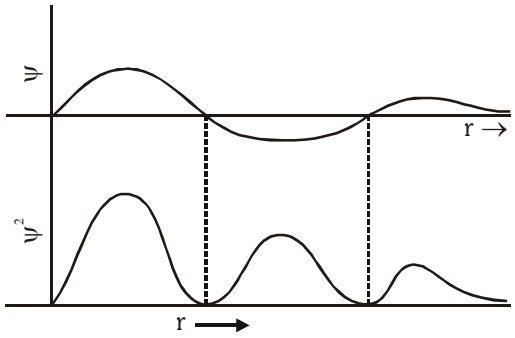


(A) $5p_z$ (B) $6d_{xy}$ (C) $6d_{x^2-y^2}$ (D) $6d_{yz}$

AS0143

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69. The wave function ψ vs radial distance (r) and square of wave function $(\psi)^2$ vs radial distance (r) graph of certain orbital is given. Identify the orbital.



(A) 3s (B) 4d (C) 3p (D) 4p

AS0144

70. The distance of spherical nodes from nucleus for the given orbital are

$$\psi_{radial} = \frac{1}{9\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2} [\sigma^2 - 4\sigma + 3] \exp(-\sigma/2)$$

where a_0 & Z are the constants and $\sigma = \frac{2Zr}{na_0}$

(A) Zero, infinity (B) $\frac{a_0}{Z}, \frac{a_0}{2Z}$

(C) $\frac{3a_0}{2Z}, \frac{9a_0}{2Z}$ (D) $\frac{a_0}{Z}, \frac{3a_0}{2Z}$

AS0145

71. The radial funtion for an orbitals is :

$$R(r) = \frac{1}{9\sqrt{6}}\left(\frac{1}{a_0}\right)^{5/2} .(4-\sigma)\ \sigma e^{-\sigma/2}; \sigma = \frac{2r}{na_0}.$$

Pick out the incorrect statement :

(A) orbital must be 3p

(B) orbital must be $3p_x$

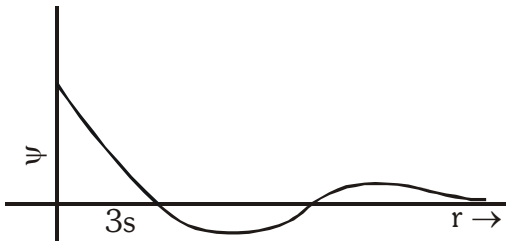
(C) the orbital has one radial node

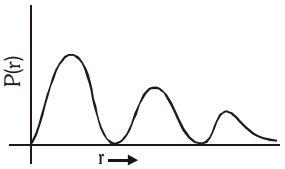
(D) the orbital must have one angular node

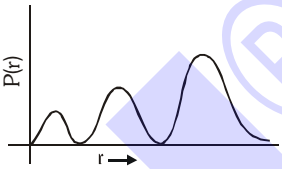
AS0146

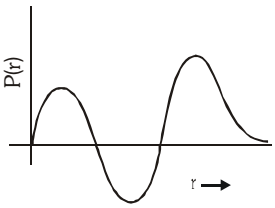
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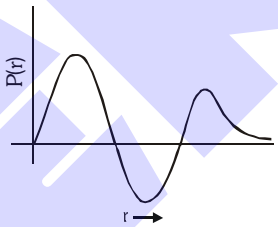
72. The wave function (ψ) verses radial distance (r) curve for certain orbital is given. Predict the shape of $\psi^2.4\pi r^2$ (radial probability distribution function) verses r graph.



(A) 

(B) 

(C) 

(D) 

AS0147

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EXERCISE # O-II

SINGLE CORRECT :

1. In a hydrogen like sample electron is in 2nd excited state, the energy of 4th state of this sample is –13.6 eV, then incorrect statement is :

(A) Atomic number of element is 4.

(B) 3 different types of spectral line will be observed if electrons make transition upto ground state from the 2nd excited state.

(C) A 25 eV photon can set free the electron from the 2nd excited state of this sample

(D) 2nd line of Balmer series of this sample has same energy value as 1st excitation energy of H-atoms.

AS0148

2. An electron in a hydrogen atom in its ground state absorbs energy equal to the ionisation energy of Li^{+2}. The wavelength of the emitted electron is:

(A) 3.32×10^{-10} m (B) 1.17 Å (C) 2.32×10^{-9} nm (D) 3.33 pm

AS0149

3. Given ΔH for the process $Li(g) \longrightarrow Li^{+3}(g) + 3e^-$ is 19800 kJ/mole & IE_1 for Li is 520 then IE_2 & IE_3 of Li are respectively (approx, value)

(A) 7505 , 11775 (B) 520 , 19280 (C) 11775 , 19280 (D) Data insufficient

AS0150

4. Consider the following nuclear reactions involving X & Y.

$$X \longrightarrow Y + {}^{4}_{2}He$$

$$Y \longrightarrow {}_{8}O^{18} + {}_{1}H^{1}$$

If both neutrons as well as protons in both the sides are conserved in nuclear reaction then moles of neutrons in 4.6 gm of X

(A) 2.4 N_A (B) 2.4 (C) 4.6 (D) 0.2 N_A

AS0151

ASSERTION AND REASON :

5. **Statement-1:** Energy emitted when an electron jump from $5 \rightarrow 2$ (energy level) is less than when an electron jump from $2 \rightarrow 1$ in all 'H' like atom.

Statement-2: The |total energy difference| between 1st & 2nd energy level is greater than that of any two energy level provided level '1' is not part of those two energy levels.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

AS0152

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6. **Statement-1** : Emitted radiations will fall in visible range when an electron jump from higher level to n = 2 in Li^{+2} ion.

Statement-2 : Balmer series radiations belong to visible range in all H-atoms.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

AS0153

MORE THAN ONE MAY BE CORRECT :

7. Which of the following could be derived from Rutherford's α-particle scattering experiment-

(A) Most of the space in the atom is empty

(B) The radius of the atom is about 10^{-10} m while that of nucleus is 10^{-15}m

(C) Electrons move in a circular path of fixed energy called orbits

(D) Electrons and the nucleus are held together by electrostatic forces of attraction.

AS0154

8. In a H-like sample electrons make transition from 5th excited state to 2nd excited state

(A) 10 different spectral lines will be emiited

(B) 6 different spectral lines will be emiited

(C) Number of lines belonging to Balmer series will be 4

(D) Number of lines belonging to paschen series will be 3

AS0155

9. The kinetic energy of photoelectron emitted on irradiating a metal surface with frequency υ is related by KE = hν – ϕ. The plots of KE vs. incidented frequency ν shows :

(A) A straight line with slope equal to Planck's constant.

(B) A straight line with intercept on x-axis equal to the product of threshold frequency and Plank's constant.

(C) A straight line with extrapolated intercept on y-axis equal to threshold energy.

(D) A straight line with intercept on x-axis equal to threshold frequency.

AS0156

10. Select the correct statement(s):

(A) All electromagnetic radiation travel with speed of light in vaccum.

(B) Energy of photon of UV light is lower than that of yellow light.

(C) He^+ and H have identical spectrum.

(D) The total energy of an electron in unielectronic specie is greater than zero

AS0157

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11. Choose the incorrect statement(s):

(A) Increasing order of wavelength is

Micro waves > Radio waves > IR waves > visible waves > UV waves

(B) The order of Bohr radius is (r_n : where n is orbit number for a given atom) $r_1 < r_2 < r_3 < r_4$

(C) The order of total energy is (E_n : where n is orbit number for a given atom) $E_1 > E_2 > E_3 > E_4$

(D) The order of velocity of electron in H, He^+, Li^+, Be^{3+} species in second Bohr orbit is

$Be^{3+} > Li^{+2} > He^+ > H$

AS0158

12. Select the correct curve(s):

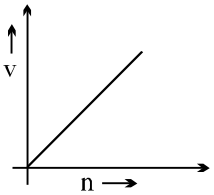
If v = velocity of electron in Bohr's orbit

r = Radius of electron in Bohr's orbit

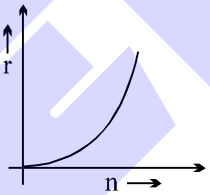
P.E. = Potential energy of electron in Bohr's orbit

K.E. = Kinetic energy of electron in Bohr's orbit.

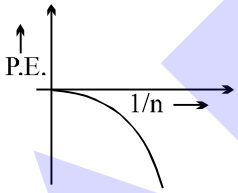
(A)



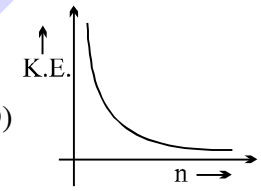
(B)



(C)



(D)



AS0159

13. Which is / are correct statement.

(A) The difference in angular momentum associated with the electron present in consecutive orbits of H-atom is $(n-1)\frac{h}{2\pi}$.

(B) Energy difference between energy levels will be changed if, P.E. at infinity assigned value other than zero.

(C) Frequency of spectral line in a H-atom is in the order of $(2 \to 1) < (3 \to 1) < (4 \to 1)$.

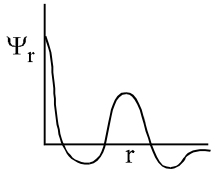
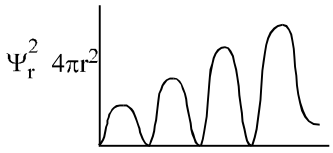
(D) On moving away from the nucleus, kinetic energy of electron decreases.

AS0160

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MATCH THE COLUMN :

14. Column I & Column II contain data on Schrondinger Wave–Mechanical model, where symbols have their usual meanings.Match the columns.

	Column I		**Column II (Type of orbital)**
(A)		(P)	4s
(B)		(Q)	$5p_x$
(C)	$\Psi(\theta, \phi) = K$ (independent of θ & ϕ)	(R)	3s
(D)	atleast one angular node is present	(S)	$6d_{xy}$

AS0161

15.

	Column-I		**Column-II**
(A)	Electron moving in 2nd orbit in He^+ ion	(P)	Radius of orbit in which electron is moving is 0.529 Å
(B)	Electron moving in 3rd orbit in H-atom	(Q)	Total energy of electron is $(-)13.6 \times 9$eV
(C)	Electron moving in 1st orbit in Li^{+2} ion	(R)	Velocity of electron is $\frac{2.188\times10^6}{3}$ m/sec
(D)	Electron moving in 2nd orbit is Be^{+3} ion	(S)	De-broglie wavelength of electron is $\sqrt{\frac{150}{13.6}}$ Å

AS0162

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COMPREHENSION :

Question No. 16 to 18 (3 questions)

The French physicist Louis de Broglie in 1924 postulated that matter, like radiation, should exhibit a dual behaviour. He proposed the following relationship between the wavelength λ of a material particle, its linear momentum p and planck constant h.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

The de Broglie relation implies that the wavelength of a particle should decreases as its velocity increases. It also implies that for a given velocity heavier particles should have shorter wavelength than lighter particles. The waves associated with particles in motion are called matter waves or de Broglie waves. These waves differ from the electromagnetic waves as they

(i) have lower velocities

(ii) have no electrical and magnetic fields and

(iii) are not emitted by the particle under consideration.

The experimental confirmation of the de Broglie relation was obtained when Davisson and Germer, in 1927, observed that a beam of electrons is diffracted by a nickel crystal. As diffraction is a characteristic property of waves, hence the beam of electron behaves as a wave, as proposed by de Broglie.

Werner Heisenberg considered the limits of how precisely we can measure properties of an electron or other microscopic particle like electron. He determined that there is a fundamental limit of how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the "Heisenberg uncertainty principle: It is impossible to determine simultaneously and precisely both the momentum and position of a particle. The product of uncertainty in the position, Δx and the uncertainty in the momentum Δ(mv) must be greater than or equal to $\frac{h}{4\pi}$. i.e.

$$\Delta x \ \Delta(mv) \geq \frac{h}{4\pi}$$

16. The correct order of wavelength of Hydrogen ($_1H^1$), Deuterium ($_1H^2$) and Tritium ($_1H^3$) moving with same kinetic energy is

(A) $\lambda_H > \lambda_D > \lambda_T$ (B) $\lambda_H = \lambda_D = \lambda_T$ (C) $\lambda_H < \lambda_D < \lambda_T$ (D) $\lambda_H < \lambda_D > \lambda_T$

AS0163

17. The transition, so that the de-Broglie wavelength of electron becomes 3 times of its initial value in He^+ ion will be

(A) $2 \rightarrow 5$ (B) $3 \rightarrow 2$ (C) $2 \rightarrow 6$ (D) $1 \rightarrow 2$

AS0164

18. If the uncertainty in velocity & position is same, then the uncertainty in momentum will be

(A) $\sqrt{\frac{hm}{4\pi}}$ (B) $m\sqrt{\frac{h}{4\pi}}$ (C) $\sqrt{\frac{h}{4\pi m}}$ (D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

AS0165

Question No. 19 to 22 (4 questions)

The only electron in the hydrogen atom resides under ordinary conditions on the first orbit. When energy is supplied, the electron moves to higher energy orbit depending on the amount of energy absorbed. When this electron returns to any of the lower orbits, it emits energy. Lyman series is formed when the electron returns to the lowest orbit while Balmer series is formed when the electron returns to second orbit. Similarly, Paschen, Brackett and Pfund series are formed when electron returns to the third, fourth and fifth orbits from higher energy orbits respectively.

Maximum number of lines produced when an electron jumps from nth level to ground level is equal to $\frac{n(n-1)}{2}$. For example, in the case of $n = 4$, number of lines produced is 6. ($4 \to 3$, $4 \to 2$, $4 \to 1$, $3 \to 2$, $3 \to 1$, $2 \to 1$). When an electron returns from n_2 to n_1 state, the number of lines in the spectrum will be equal to

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

If the electron comes back from energy level having energy E_2 to energy level having energy E_1, then the difference may be expressed in terms of energy of photon as:

$$E_2 - E_1 = \Delta E \ , \ \lambda = \frac{hc}{\Delta E}$$

Since h and c are constants, ΔE corresponds to definite energy; thus each transition from one energy level to another will produce a light of definite wavelength. This is actually observed as a line in the spectrum of hydrogen atom.

Wave number of line is given by the formula $\overline{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$.

where R is a Rydberg's constant ($R = 1.1 \times 10^7\ m^{-1}$)

19. The energy of photon emitted corresponding to transition n = 3 to n = 1 is [$h = 6 \times 10^{-34}$ J-sec.]

(A) 1.76×10^{-18} J (B) 1.98×10^{-18} J (C) 1.76×10^{-17} J (D) None of these

AS0166

20. In a collection of H-atom, electrons make transition from 5th excited state to 2nd excited state then maximum number of different types of photons observed are

(A) 3 (B) 4 (C) 6 (D) 15

AS0167

21. The difference in the wavelength of the 1st line of Lyman series and 2nd line of Balmer series in a hydrogen atom is

(A) $\frac{9}{2R}$ (B) $\frac{4}{R}$ (C) $\frac{88}{15R}$ (D) None

AS0168

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22. The wave number of electromagnetic radiation emitted during the transition of electron in between two levels of Li^{2+} ion whose principal quantum numbers sum is 4 and difference is 2 is

(A) 3.5 R (B) 4 R (C) 8 R (D) $\frac{8}{9}R$

AS0169

1 TABLE (3Q)

Column-I	Column-II	Column-III
(A) Energy of electron in H-like atomic orbital	(P) Principal quantum number(n)	(I) Radial function [R(r)]
(B) Orientation and shape of H-like atomic orbital	(Q) Azimuthal quantum number (l)	(II) Angular function
(C) Magnitude & direction of spin angular momentum	(R) Magnetic quantum number (m)	(III) Wave function [Ψ]
(D) Probability density of electron in s-orbital of H-like atom/ion	(S) Spin quantum number(s)	(IV) Cannot be determined from Schrodinger equation

Assume no external electric or magnetic fields in Q.23 to Q.25

23. Which of the following option is incorrectly matched -

(A) B – Q – II (B) B – R – II (C) B – P – I (D) B – Q – III

AS0170

24. Which option is correctly matched ?

(A) C – S – III (B) D – R – III (C) A – P – IV (D) C – S – IV

AS0170

25. If option (A) is considered in presence of external electric or magnetic field then which option is incorrect -

(A) A – P – III (B) A – Q – III (C) A – R – III (D) A – S – I

AS0170

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EXERCISE # J-MAINS

1. The wave number of the first emission line in the Balmer series of H-Spectrum is : (R = Rydberg constant) : **[JEE-Main(online) 2013]**

(1) $\frac{3}{4}R$ (2) $\frac{9}{400}R$ (3) $\frac{5}{36}R$ (4) $\frac{7}{6}R$

AS0173

2. For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship? **[JEE-Main(online) 2014]**

(1) a dust particle (2) an electron (3) a proton (4) an α-particle.

AS0175

3. If the binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is : **[JEE-Main(online) 2014]**

(1) 13.6 eV (2) 30.6 eV (3) 122.4 eV (4) 3.4 eV

AS0176

4. If λ_0 and λ be the threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is **[JEE-Main(online) 2014]**

(1) $\sqrt{\frac{2hc}{m}\left(\frac{\lambda_0-\lambda}{\lambda\lambda_0}\right)}$ (2) $\sqrt{\frac{2h}{m}\left(\frac{1}{\lambda_0}-\frac{1}{\lambda}\right)}$ (3) $\sqrt{\frac{2h}{m}(\lambda_0-\lambda)}$ (4) $\sqrt{\frac{2hc}{m}(\lambda_0-\lambda)}$

AS0178

5. At temperature T, the average kinetic energy of any particle is $\frac{3}{2}$ kT- The de Broglie wavelength follows the order : **[JEE-Main(online) 2015]**

(1) Visible photon > Thermal electron > Thermal neutron

(2) Thermal proton > Thermal electron > Visible photon

(3) Visible photon > Thermal neutron > Thermal electron

(4) Thermal proton > Visible photon > Thermal electron

AS0181

6. A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V volts. If e and m are charge and mass of an electron respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by : **[JEE-Main(online) 2016]**

(1) $\sqrt{2meV}$ (2) meV (3) 2meV (4) $\sqrt{meV}$

AS0182

7. If the shortest wavelength in Lyman series of hydrogen atom is A, then the longest wavelength in Paschen series of He^+ is : **[JEE-Main(online) 2017]**

(1) $\frac{36A}{5}$ (2) $\frac{9A}{5}$ (3) $\frac{36A}{7}$ (4) $\frac{5A}{9}$

AS0184

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8. The electron in the hydrogen atom undergoes transition from higher orbitals to orbital of radius 211.6 pm. This transition is associated with:- **[JEE-Main(online) 2017]**

(1) Brackett series (2) Balmer series (3) Lyman series (4) Paschen series

AS0185

9. Ejection of the photoelectron from metal in the photoelectric effect experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function of the metal is :

(1) 5 eV (2) 4 eV (3) 5.5 eV (4) 4.5 eV **[JEE-Main(online) 2018]**

AS0187

10. Which of the following statements is false ? **[JEE-Main(online) 2018]**

(1) Photon has momentum as well as wavelength.

(2) Splitting of spectral lines in electrical field is called Stark effect.

(3) Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.

(4) Rydberg constant has unit of energy.

AS0188

11. If p is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength λ, then for 1.5 p momentum of the photoelectron, the wavelength of the light should be: **[JEE-Main(online) 2019]**

(Assume kinetic energy of ejected photoelectron to be very high in comparison to work function)

(1) $\frac{1}{2}\lambda$ (2) $\frac{3}{4}\lambda$ (3) $\frac{2}{3}\lambda$ (4) $\frac{4}{9}\lambda$

AS0189

12. For any given series of spectral lines of atomic hydrogen, let $\Delta\bar{\nu} = \bar{\nu}_{max} - \bar{\nu}_{min}$ be the difference in maximum and minimum frequencies in cm^{-1}. The ratio $\Delta\bar{\nu}_{Lyman} / \Delta\bar{\nu}_{Balmer}$ is :

(1) 27 : 5 (2) 4 : 1 **[JEE-Main(online) 2019]**

(3) 5 : 4 (4) 9 : 4

AS0191

13. The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are: **[JEE-Main(online) 2019]**

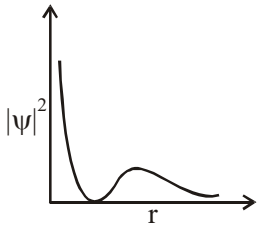
(1) Paschen and P fund (2) Lyman and Paschen

(3) Brackett and Piund (4) Balmer and Brackett

AS0192

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14. The graph betweeen $|\psi|^2$ and r(radial distance) is shown below. This represents :-



(1) 3s orbital (2) 1s orbital **[JEE-Main(online) 2019]**

(3) 2p orbital (4) 2s orbital

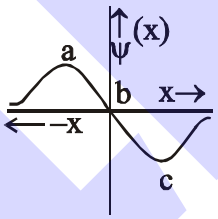
AS0193

15. Among the following, the energy of 2s orbital is lowest in : **[JEE-Main(online) 2019]**

(1) K (2) Na (3) Li (4) H

AS0194

16. The electrons are more likely to be found : **[JEE-Main(online) 2019]**



(1) in the region a and b (2) in the region a and c

(3) only in the region c (4) only in the region a

AS0195

17. If the de Broglie wavelength of the electron in n^{th} Bohr orbit in a hydrogenic atom is equal to $1.5\ \pi a_0$(a_0 is Bohr radius), then the value of n/z is : **[JEE-Main(online) 2019]**

(1) 1.0 (2) 0.75 (3) 0.40 (4) 1.50

AS0197

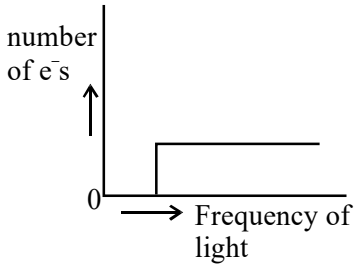
18. The de Broglie wavelength (λ) associated with a photoelectron varies with the frequency (ν) of the incident radiation as, [ν_0 is thershold frequency] : **[JEE-Main(online) 2019]**

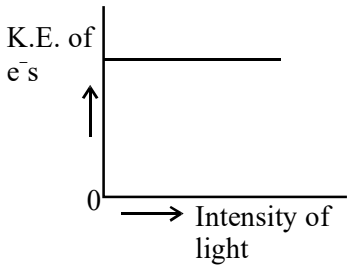
(1) $\lambda \propto \frac{1}{(\nu-\nu_0)^{\frac{3}{2}}}$ (2) $\lambda \propto \frac{1}{(\nu-\nu_0)^{\frac{1}{2}}}$ (3) $\lambda \propto \frac{1}{(\nu-\nu_0)^{\frac{1}{4}}}$ (4) $\lambda \propto \frac{1}{(\nu-\nu_0)}$

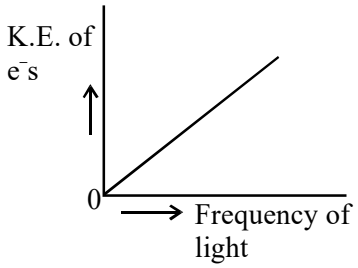
AS0200

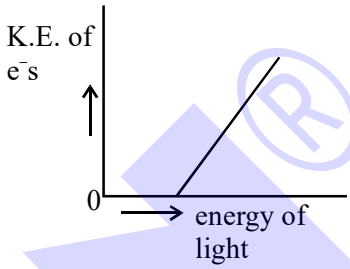
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19. Which of the graphs shown below does not represent the relationship between incident light and the electron ejected form metal surface ? **[JEE-Main(online) 2019]**

(1)


(2)


(3)


(4)


AS0202

20. Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect? (The Bohr radius is represented by a_0) **[JEE-Main(online) 2019]**

(1) The electron can be found at a distance $2a_0$ from the nucleus

(2) The probability density of finding the electron is maximum at the nucleus.

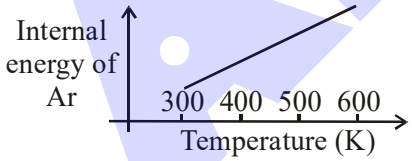
(3) The magnitude of potential energy is double that of its kinetic energy on an average.

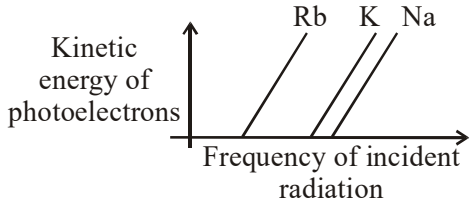
(4) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.

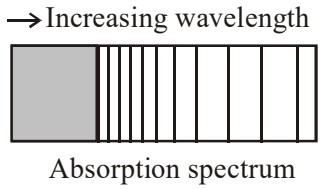
AS0232

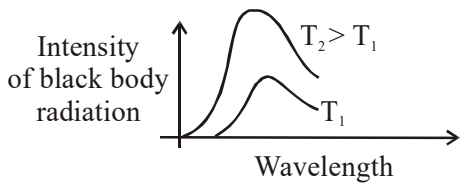
21. The figure that is not a direct manifestation of the quantum nature of atoms is :

[JEE-Main(online) 2020]

(1)


(2)


(3)


(4)


AS0233

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22. The correct statement about probability density (except at infinite distance from nucleus) is : **[JEE-Main(online) 2020]**

(1) It can be negative for 2p orbital (2) It can be zero for 3p orbital
(3) It can be zero for 1s orbital (4) It can never be zero for 2s orbital

AS0234

23. The difference between the radii of 3rd and 4th orbits of Li^{2+} is ΔR_1. The difference between the radii of 3rd and 4th orbits of He^+ is ΔR_2. Ratio $\Delta R_1 : \Delta R_2$ is : **[JEE-Main(online) 2020]**

(1) 8 : 3 (2) 3 : 2 (3) 3 : 8 (4) 2 : 3

AS0235

24. According to Bohr's atomic theory :- **[JEE-Main(online) 2021]**

(A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$.

(B) The product of velocity (v) of electron and principal quantum number (n), 'vn' $\propto Z^2$.

(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.

(D) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$.

Choose the most appropriate answer from the options given below :

(1) (C) Only (2) (A) Only (3) (A), (C) and (D) only (4) (A) and (D) only

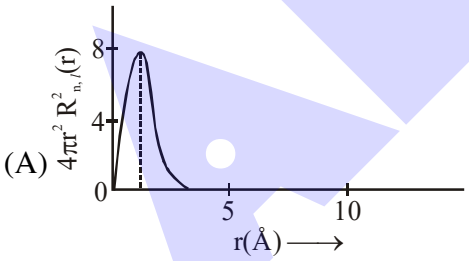
AS0236

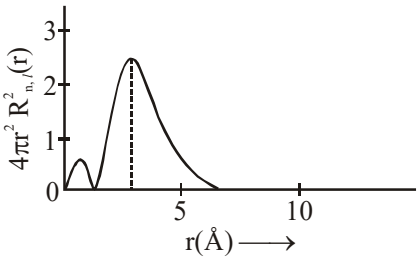
25. A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_P denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{Li}}{\lambda_P}$ is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer) (Mass of Li^{3+} = 8.3 mass of proton)

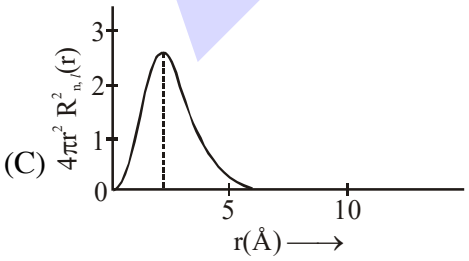
[JEE-Main(online) 2021]

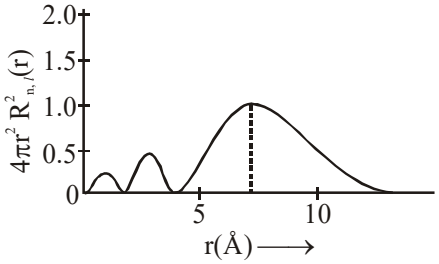
AS0237

26. The plots of radial distribution functions for various orbitals of hydrogen atom against 'r' are given below: **[JEE-Main(online) 2021]**

(A) 

(B) 

(C) 

(D) 

The correct plot for 3s orbital is:

(1) (B) (2) (A) (3) (D) (4) (C)

AS0238

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27. A certain orbital has n = 4 and $m_L = -3$. The number of radial nodes in this orbital is _____. (Round off to the Nearest Integer). **[JEE-Main(online) 2021]**

AS0239

28. Given below are two statements : **[JEE-Main(online) 2021]**

Statement I : Bohr's theory accounts for the stability and line spectrum of Li^+ ion.

Statement II : Bohr's theory was unable to explain the splitting of spectral lines in the presence of a magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are true. (2) Statement I is false but statement II is true.

(3) Both statement I and statement II are false. (4) Statement I is true but statement II is false.

AS0240

29. A certain orbital has no angular nodes and two radial nodes. The orbital is : **[JEE-Main(online) 2021]**

(1) 2s (2) 3s (3) 3p (4) 2p

AS0241

30. If the Thomson model of the atom was correct, then the result of Rutherford's gold foil experiment would have been : **[JEE-Main(online) 2021]**

(1) All of the α-particles pass through the gold foil without decrease in speed.

(2) α-Particles are deflected over a wide range of angles.

(3) All α-particles get bounced back by 180°

(4) α-Particles pass through the gold foil deflected by small angles and with reduced speed.

AS0242

31. Given below are two statements : **[JEE-Main(online) 2021]**

Statement I : Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.

Statement II : Bohr's model of hydrogen atom contradicts Heisenberg's uncertainty principle.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is false but statement II is true. (2) Statement I is true but statement II is false.

(3) Both statement I and statement II are false. (4) Both statement I and statement II are true.

AS0243

32. Given below are two statements. **[JEE-Main(online) 2021]**

Statement I : According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.

Statement II: According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are false (2) Both Statement I and Statement II are true

(3) Statement I is false but Statement II is true (4) Statement I is true but Statement II is false

AS0244

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33. The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1 mW and wavelength of 1000 nm, in 0.1 second is $x \times 10^{13}$. The value of x is _____. (Nearest integer)
($h = 6.63 \times 10^{-34}$ Js, $c = 3.00 \times 108$ ms^{-1}) **[JEE-Main(online) 2021]**

AS0245

34. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{xma_0^2}$. The value of 10x is _____ . (a_0 is radius of Bohr's orbit) (Nearest integer) [Given : $\pi = 3.14$]

AS0246

[JEE-Main(online) 2021]

35. The energy of one mole of photons of radiation of wavelength 300 nm is
(Given : $h = 6.63 \times 10^{-34}$ Js, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $c = 3 \times 10^8$ ms^{-1}) **[JEE-Main(online) 2022]**

(A) 235 kJ mol^{-1} (B) 325 kJ mol^{-1} (C) 399 kJ mol^{-1} (D) 435 kJ mol^{-1}

AS0247

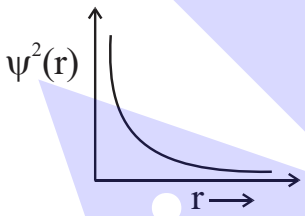
36. The number of radial and angular nodes in 4d orbital are. respectively

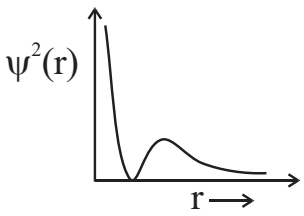
[JEE-Main(online) 2022]

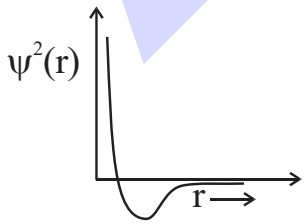
(A) 1 and 2 (B) 3 and 2 (C) 1 and 0 (D) 2 and 1

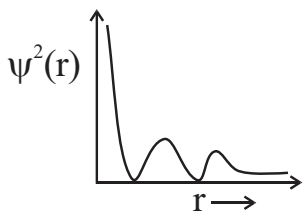
AS0248

37. Which of the following is the correct plot for the probability density $\psi^2(r)$ as a function of distance 'r' of the electron form the nucleus for 2s orbital? **[JEE-Main(online) 2022]**

(A)


(B)


(C)


(D)


AS0249

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38. When the excited electron of a H atom from n = 5 drops to the ground state, the maximum number of emission lines observed are _____ **[JEE-Main(online) 2022]**

AS0250

39. The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is_____. (Nearest Integer)

(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

[JEE-Main(online) 2022]

AS0251

40. Identify the incorrect statement from the following. **[JEE-Main(online) 2022]**

(A) A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.

(B) An orbital is the one electron wave function (ψ) in an atom.

(C) The existence of Bohr's orbits is supported by hydrogen spectrum.

(D) Atomic orbital is characterised by the quantum numbers n and l only

AS0252

41. The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_0$ (Where a_0 = Bohr radius 52.9 pm) is _____km s^{-1}.

(Given : Mass of electron = 9.1×10^{-31} kg, Planck's constant h = 6.63×10^{-34} Js)

[JEE-Main(online) 2022]

AS0253

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EXERCISE # J-ADVANCED

Paragraph for questions 1 to 3

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom. **[JEE 2010]**

1. The state S_1 is :-

(A) 1s (B) 2s (C) 2p (D) 3s

AS0209

2. Energy of the state S_1 in units of the hydrogen atom ground state energy is :-

(A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

AS0210

3. The orbital angular momentum quantum number of the state S_2 is :-

(A) 0 (B) 1 (C) 2 (D) 3

AS0211

4. The maximum number of electrons that can have principal quantum number, n=3, and spin quantum number, $m_s = -1/2$, is **[JEE 2011]**

AS0212

5. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is : :- **[JEE 2011]**

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ(eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

AS0213

6. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] **[JEE 2012]**

(A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$ (C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{64\pi^2 m a_0^2}$

AS0214

7. The atomic masses of He and Ne are 4 and 20 a.m.u. respectively. The value of the de Broglie wavelength of He gas at –73 °C is "M" times that of the de Broglie wavelength of Ne at 727 °C. M is. **[JEE 2013]**

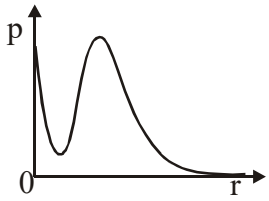
AS0215

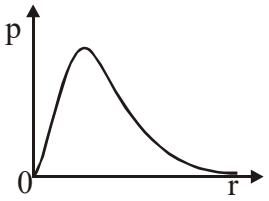
8. In an atom, the total number of electrons having quantum numbers n = 4, $|m_\ell| = 1$ and $m_s = -\frac{1}{2}$ is **[JEE 2014]**

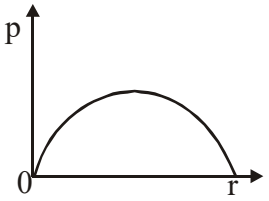
AS0216

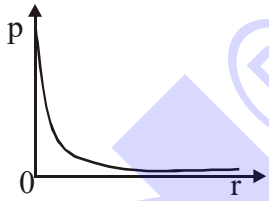
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9. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr, at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is - **[JEE 2016]**

(A) 

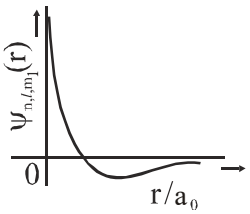
(B) 

(C) 

(D) 

AS0217

Answer Q.10, Q.11 and Q.12 by appropriately matching the information given in the three columns of the following table. **[JEE 2017]**

Column-1	**Column-2**	**Column-3**
(I) 1s orbital	(i) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$	(P) 
(II) 2s orbital	(ii) One radial node	(Q) Probability density at nucleus $\propto \frac{1}{a_0^3}$
(III) $2p_z$ orbital	(iii) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{2a_0}\right)} \cos\theta$	(R) Probability density is maximum at nucleus
(IV) $3d_z^2$ orbital	(iv) xy - plane is a nodal plane	(S) Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from n = 2 state to n = 6 state

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10. For the given orbital in column 1, the only **CORRECT** combination for any hydrogen - like species is :

(A) (IV) (iv) (R) (B) (II) (ii) (P) (C) (III) (iii) (P) (D) (I) (ii) (S)

AS0218

11. For He^+ ion, the only **INCORRECT** combination is

(A) (II) (ii) (Q) (B) (I) (i) (S) (C) (I) (i) (R) (D) (I) (iii) (R)

AS0219

12. For hydrogen atom, the only **CORRECT** combination is

(A) (I) (iv) (R) (B) (I) (i) (P) (C) (II) (i) (Q) (D) (I) (i) (S)

AS0220

13. The ground state energy of hydrogen atom is –13.6 eV. Consider an electronic state Ψ of He^+ whose energy, azimuthal quantum number and magnetic quantum number are –3.4 eV, 2 and 0 respectively. Which of the following statement(s) is(are) true for the state Ψ? **[JEE 2019]**

(1) It has 2 angular nodes

(2) It has 3 radial nodes

(3) It is a 4d state

(4) The nuclear charge experienced by the electron in this state is less than 2e, where e is the magnitude of the electronic charge.

AS0221

Question No. 14 to 15 (2 questions)

Answer the following by appropriately matching the lists based on the information given in the paragraph **[JEE 2019]**

Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following List-I contains some quantities for the nth orbit of the atom and List-II contains options showing how they depend on n.

	List-I	**List-II**
(I)	Radius of the nth orbit	**(P)** $\propto n^{-2}$
(II)	Angular momentum of the electron in the nth orbit	**(Q)** $\propto n^{-1}$
(III)	Kinetic energy of the electron in the nth orbit	**(R)** $\propto n^{0}$
(IV)	Potential energy of the electron in the nth orbit	**(S)** $\propto n^{1}$
		(T) $\propto n^{2}$
		(U) $\propto n^{1/2}$

14. Which of the following options has the correct combination considering List-I and List-II?

(1) (II), (R) (2) (I), (P) (3) (I), (T) (4) (II), (Q)

AS0222

15. Which of the following options has the correct combination considering List-I and List-II?

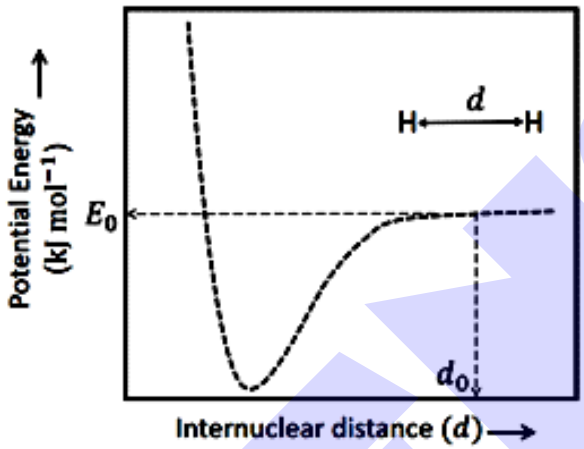
(1) (III), (S) (2) (IV), (Q) (3) (IV), (U) (4) (III), (P)

AS0223

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16. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in the figure) in kJ mol^{-1}, for d = d_0 at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart.

Use Avogadro constant as 6.023×10^{23} mol^{-1}. **[JEE 2020]**



AS0254

17. Consider a helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s-1) of He atom after the photon absorption is ___.

(Assume: Momentum is conserved when photon is absorbed.

Use: Planck constant = 6.6×10^{-34} J s, Avogadro number = 6×10^{23} mol^{-1},

Molar mass of He = 4 g mol^{-1}) **[JEE 2021]**

AS0255

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ANSWER KEY

EXERCISE # S-I

1.	1.00	2. 9.00	3. $n < \alpha < p < e$	4. 6.00
5.	1.60	6. 8.00	7. 2.30	8. 0.50
9.	1.28	10. 1.44	11. 150.00	12. 150.00
13.	0.20	14. 7.50	15. 4.96 or 4.97	16. 10.00
17.	1.00	18. 0.50	19. 620.00	20. 3.00
21.	150.13 or 150.14	22. 1.00	23. 50.00	24. 3.06
25.	0.41	26. 0.26	27. 4.00	28. 5.40 to 5.50
29.	(i) 2.00 (ii) 9.60 to 9.80		30. 8.20 to 8.40	31. 65.00 to 67.00
32.	0.85	33. 2.10 or 2.11	34. 2.00	35. 5280.00
36.	–2.18	37. (i) 5.00 (ii) 340.00		38. 3.00
39.	6566.40; 1216.00; 1026.00		40. 2.00	41. 114.00
42.	(i) 4.00 ; (ii) 0.19	43. 3.00	44. 4.00	45. 3.09
46.	3.00	47. 8.00	48. 4.00	49. 36.81
50	10.00	51. 0.08	52. 0.11	53. 0.17
54.	8.00	56. 3.75	57. 3.00	58. 2.00
59.	0.63	60. 4.00		

EXERCISE # S-II

1.	20.00	2. $E = \frac{n^6 h^6}{384\, m^3\, K^2\, e^4\, \pi^6}$	
3.	(i) 1, (ii) 2, (iii) 3, (iv) 3	4. – 435.20	
5.	90.00	6. 91.00	
7.	(a) Z = 3 ; (b) $n_1 = 12$, $n_2 = 15$; (c) 0.08	8. 9.68 or 9.69	
9.	972.54 or 972.55	10. 300.00, 0.26	
11.	(a) 75.00; (b) 1.41	12. 0.50	
13.	0.36	14. 1240.00	
15.	300303		

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EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	C	C	A	D	D	D	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	B	B	B	B	B	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	A	B	B	A	C	C	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	B	D	C	B	C	D	A	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	D	B	C	C	A	D	B	A	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	A	C	C	A	A	A	B	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	D	B	C	B	C	C	D	D	C
Que.	71	72								
Ans.	B	B								

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	B	A	D	A,B,D	B,D	A,D	A
Que.	11	12	13	14						
Ans.	A,C	B,C,D	C,D	A-P	B-P,Q,S	C-P,R	D-Q,S			
Que.	15				16	17	18	19	20	21
Ans.	A-S	B-R	C-Q	D-P	A	C	A	A	C	B
Que.	22	23	24	25						
Ans.	C	C	D	D						

EXERCISE # J-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	2	1	1	1	3	2	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	2	4	1	2	2	2	3	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	4	4	2	3	0	2	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	3	50	3155	C	A	B	4	1758	D
Que.	41									
Ans.	548									

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EXERCISE # J-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	9	4	C	5	6	B	B
Que.	11	12	13	14	15	16				17
Ans.	D	D	1,3	3	4	–2640.00 to –2620.00		or –5280.00 to –5240.00		30

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